

MAXWELL KORTEBEIN

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PROFESSIONAL SUMMARY

Graduate student in Geographic Information Systems with a background in data science and international education. Skilled in Python, ArcGIS, and spatial analysis, with a passion for transforming complex data into actionable insights for public and environmental benefit.

EDUCATION

University of Minnesota – Twin Cities

Minneapolis, MN

Masters in Geographic Information Systems (GIS)

Expected: May 2027

- ❖ **Relevant Courses:** Principles of Cartography, Intermediate Spatial Data Science with ArcGIS, GIS Project Management
- ❖ **GPA:** 4.0

University of Wisconsin - Madison

Madison, WI

B.S. Data Science

September 2019 – May 2023

- ❖ **Relevant Courses:** Data Programming II, Data Modelling II, Big Data Systems, Introduction to Artificial Intelligence
- ❖ **GPA:** 3.45

EXPERIENCE

University of Minnesota

Minneapolis, Minnesota

Graduate Teaching Assistant – Principles of Geocomputing

September 2025 - Present

- ❖ Assist both undergraduate and graduate students in learning Python with an emphasis on geospatial use.
- ❖ Responsible for grading labs and exercises; hosting office hours; assisting students; and redesigning labs

University of Minnesota

Minneapolis, Minnesota

Transportation Intern

September 2025 - Present

- ❖ Assist in maintaining records of public infrastructure.
- ❖ Conduct spatial analyses on bike data collected from surveys to understand ridership patterns.

Chiba Board of Education

Chiba, Japan

Assistant Language Teacher

August 2023 – August 2025

- ❖ Taught English and science to over 320 high school students, focusing on making complex scientific topics accessible and engaging.
- ❖ Mentored 20 students over two semesters in preparation for international science presentations during a one-week exchange program in Taiwan.

PROJECTS

Metro Transit Route Analysis

Minneapolis, MN

2025 – Present

- ❖ Analyzed demographics, POI, and infrastructure data, cleaning and joining features when necessary
- ❖ Created rasters showing the suitability of public transportation around the Twin Cities metro area, utilizing Python and ArcGIS Pro.
- ❖ Created optimal paths for new and existing Metro Transit routes using hot spot analysis.
- ❖ Created an interactive Esri Storymap to explain my analysis and the results from it.

North Country Trail Association - Consulting Project

Minneapolis, MN

September 2025 – December 2025

- ❖ Worked with the North Country Trail Association to reroute the current North Country Trail in Minnesota.
- ❖ As part of a large group, we analyzed parcel and POI data to create suitability hotspots for where new trails could be placed.
- ❖ Created interactive maps and visualizations in ArcGIS Pro and Experience Builder for the North Country Trail Association, with the goal of sharing our analysis clearly and efficiently..

Metro Transit Air Quality Analysis

Minneapolis, MN

December 2025 – Present 2025

- ❖ Currently assisting professor Ying Song with her research into the adverse health effects of air quality surrounding buses in the Twin Cities metro area.
- ❖ Our goal is to find the measurements of CO2 and PM2.5 both inside and outside of the bus at different places and times during the day, with the target concern being bus drivers' long-term health effects.
- ❖ Created a Python script which gathers data from the air quality sensors API and transforms with into clear metrics and maps to assist with visualization.

TECHNICAL SKILLS

Programming & Tools: Python, SQL, ArcGIS Pro, QGIS, Git, Docker, GeoJSON, Data Visualization

Analytical Skills: Spatial Analysis, Data Wrangling, Statistical Modeling, Public Data Interpretation

Soft Skills: Communication, Critical Thinking, Public Speaking, Mentorship

LEADERSHIP & ACTIVITIES

Tabling Event - U of M 2025 Spatial Forum

Minneapolis, MN

November 2025

- ❖ Hosted a tabling event in which I displayed my Metro Transit Route Analysis to a variety of people from different backgrounds.
- ❖ Attended a lecture series about different aspects of geospatial data science and expanded my network.

ESS Club

Chiba, Japan

Club Leader

2023 – 2025

- ❖ Organized weekly English-speaking meetups and cultural exchange events.
- ❖ Mentored club members in English language learning and presentation skills.

Basketball Intramural Club

Chiba, Japan

Club Member

2023 – 2025

- ❖ Participate in weekly basketball practices and games, promoting teamwork, discipline, and physical wellness.
- ❖ Support club event organization and help integrate new members into the group.

ADDITIONAL INFORMATION

Languages: English (native), Japanese (intermediate)

Interests: Basketball, Reading, Music, Piano, Language Learning